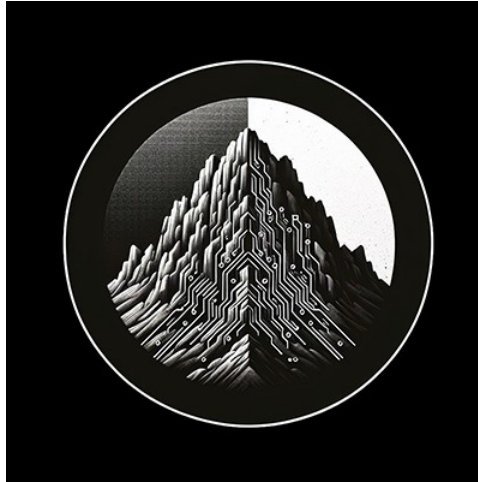




# RTL Design Sherpa

## STREAM Characterization Monbus Compression Report

June 2026

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# Readme

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## Monbus Compression — Characterization (all-fixes bitstream)

Companion to `../perf/`. Where the perf report measures how hard the DMA drives the bus, this measures how well the **monbus compressor** + **half-beat packer** shrink the monitor-event stream on real captured traffic.

**Bitstream:** all-fixes build (`USE_MON_COMPRESSION = USE_MON_HALFBEAT = USE_MON_CAM_PIPELINE = 1`), WNS +0.048 ns @ 100 MHz, xc7a100t. **Primary data:** the committed encoder CAM logic replayed **bit-exact and offline** on the real 682-record on-board capture (`../compression_dataset/desc_axi_16desc_8ch_1MB.json`). The on-chip packer is verified bit-identical to this model in cosim (`val/amba/test_monbus_halfbeat.py`), so the offline reduction is what the FPGA emits for the same record stream — it is the codec, not an estimate.

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### 1. What compression means here

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Each monitor record is a 128-bit packet + 64-bit timestamp. A raw (Tier-0) escape costs **3 × 64-bit beats**. The codec removes that two ways:

1. **Tier-1 templating** (64-bit slot): a CAM-hit record collapses to **1 beat**.  
Structural ceiling  $1 - 1/3 = 66.7 \%$ .
2. **Half-beat packing** (32-bit slot): two compressible records share one beat (0.5 beat/record), raising the ceiling to  $1 - 0.5/3 = 83.3 \%$ .

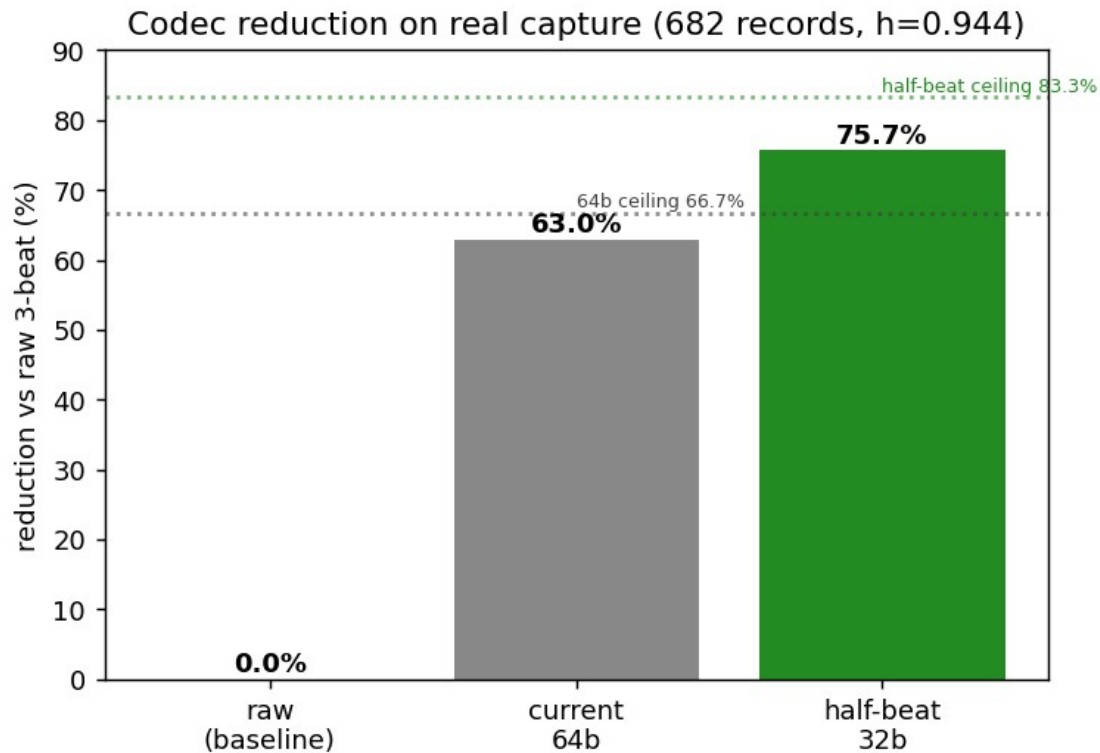
Realized reduction is dominated by the Tier-1 hit rate  $h$ , not slot size:

|                                          |                 |                |
|------------------------------------------|-----------------|----------------|
| $\text{reduction} = h \cdot (1 - b_t/3)$ |                 |                |
| current ( $b_t = 1.0$ ):                 | $h \cdot 0.667$ | ceiling 66.7 % |
| half-beat ( $b_t = 0.5$ ):               | $h \cdot 0.833$ | ceiling 83.3 % |

So 80 % needs  $h \approx 0.96$ . The point of measuring on real traffic is to find the actual  $h$  and the fraction of hits that fit a 32-bit half-slot.

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## 2. Headline (real capture, 682 records, h = 0.944)



### codec comparison

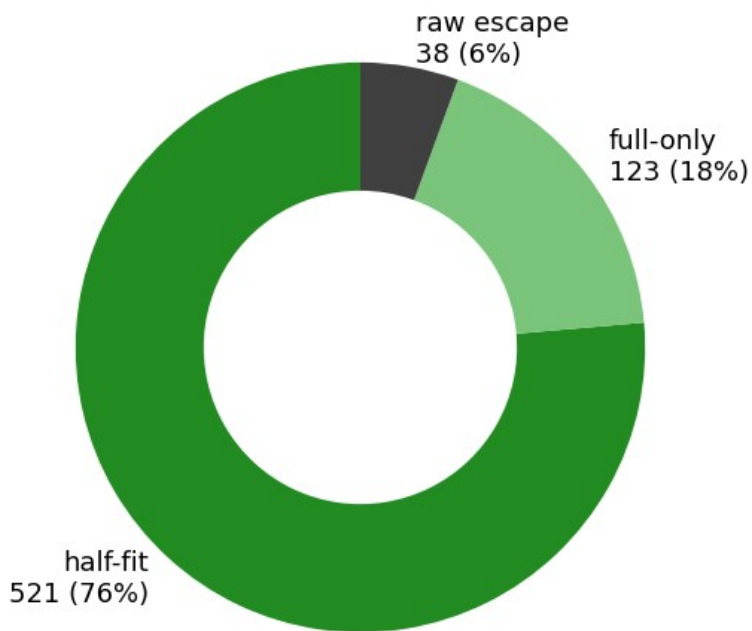
| codec                   | beats      | reduction     |
|-------------------------|------------|---------------|
| raw baseline            | 2046       | —             |
| current 64-bit          | 758        | 63.0 %        |
| <b>half-beat 32-bit</b> | <b>498</b> | <b>75.7 %</b> |

Half-beat packing turns the 63.0 % codec into 75.7 % on this trace — a **12.7-percentage-point** gain, **260 fewer beats** drained over UART for identical monitor coverage.

Why 75.7 % and not the 83.3 % ceiling:  $h = 0.944$  caps half-beat at  $0.944 \times 0.833 = 78.6 \%$ , and of the 644 hits, 123 fit only a full 64-bit beat (event\_data or  $\Delta t$ s too wide for a 23-bit half-payload), plus lone-half round-up. The gap to ceiling is real-traffic structure, not codec overhead.

### 3. Per-record classification

Per-record classification — 682 records



#### *tier composition*

| class                                    | records | share | cost     |
|------------------------------------------|---------|-------|----------|
| half-fit (Tier-1, 32-bit slot)           | 521     | 76 %  | 0.5 beat |
| full-only (Tier-1, 64-bit slot)          | 123     | 18 %  | 1 beat   |
| raw escape (Tier-0, CAM miss / overflow) | 38      | 6 %   | 3 beats  |

Three quarters of records pack into half-slots. The 18 % full-only records are Tier-1 hits whose `event_data/dts` exceed the 23-bit half-payload; the 6 % raw escapes are first-occurrence CAM misses (one per distinct template) plus a handful of `dts` overflows.

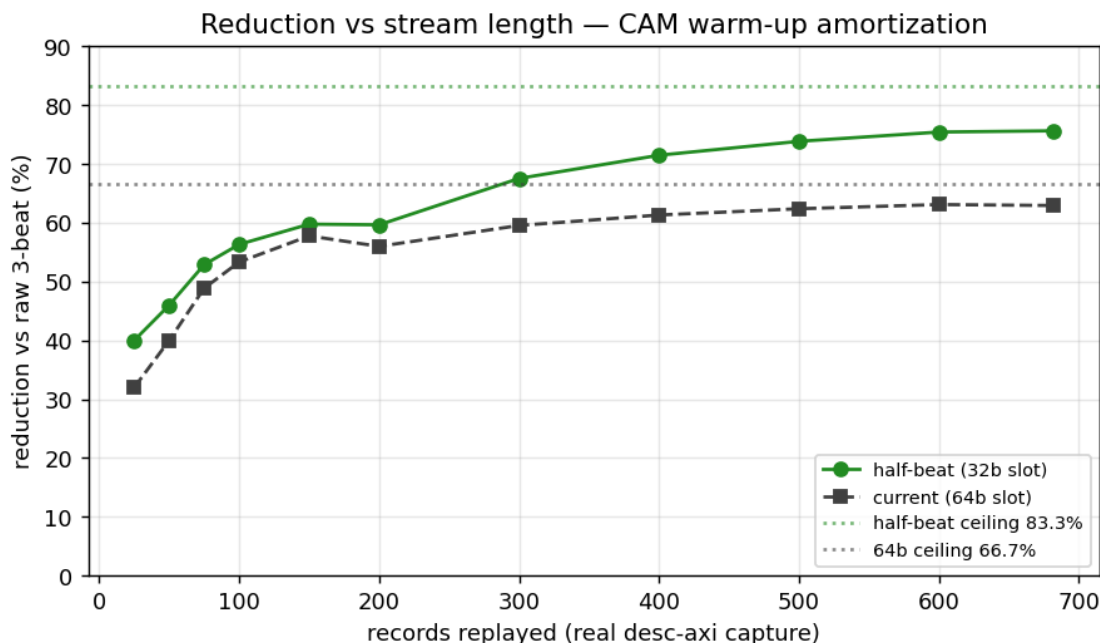
## 4. Half-beat wire format

A half-pair beat is  $\{\text{tag}[63:60]=0x4, \text{slotA}[59:30], \text{slotB}[29:0]\}$  — two 30-bit slots, each  $\text{sub}[1:0] + \text{idx}[4:0] + 23\text{-bit payload}$ :

```
HALF-A (absolute):  sub=1 | idx[4:0] | delta_ts[9:0] |  
event_data[12:0]  
HALF-C (delta):    sub=2 | idx[4:0] | delta_ts[9:0] | ed_delta[11:0]  
signed  
NOP (lone pad):    sub=0
```

$\text{delta\_ts} < 1024 \text{ cyc}$  ( $\sim 10.2 \mu\text{s}$  @ 100 MHz) for either form; HALF-A carries  $\text{event\_data} < 2^{13}$ , HALF-C a signed  $\text{ed\_delta} | \cdot | < 2^{12}$  for counters/sequential addresses. Tier-1 records that fit neither stay a full 64-bit beat; a lone trailing half rounds up to a full beat. Widths locked in `host/monbus_halfbeat_model.py` (lines 50–62), mirrored bit-exact by `rtl/amba/shared/monbus_halfbeat_packer.sv`.

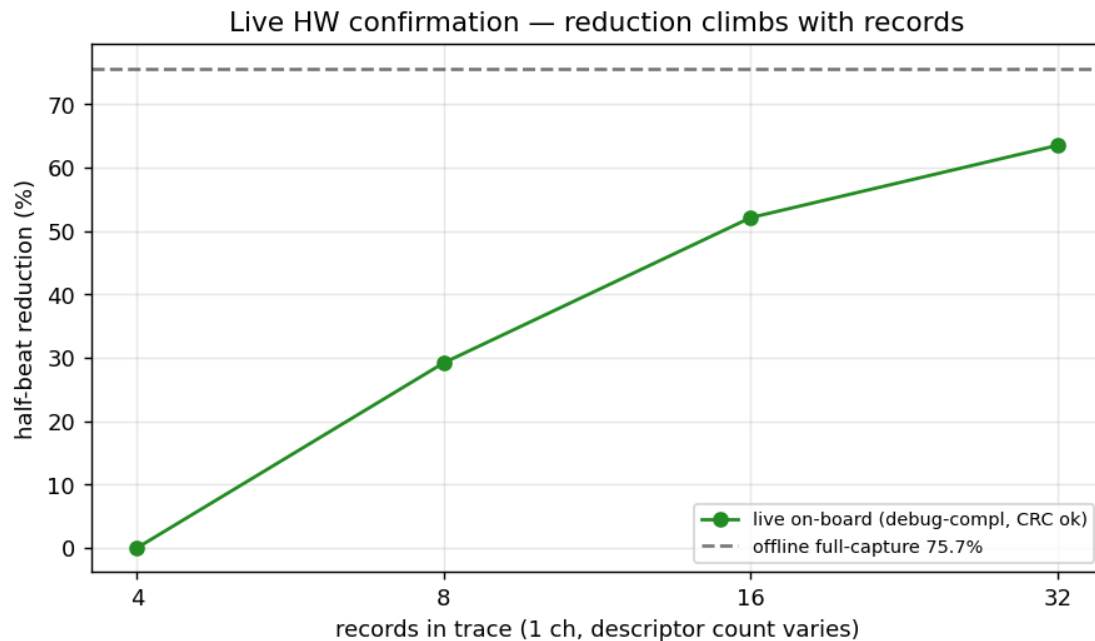
## 5. Reduction vs stream length — CAM warm-up



### *reduction vs length*

Replaying prefixes of the real capture shows the dominant real-world effect: **warm-up amortization**. Each distinct template costs one 3-beat raw escape on first sight; until those are paid off they dominate a short stream. As records accumulate the escapes amortize and reduction climbs toward the steady 75.7 %. This is why short captures under-report compression and the full-stream number is the representative one.

## 6. Live on-board confirmation



### live confirmation

The same codec is synthesized into the bitstream. `hb_measure.py` runs the compressed workload, drains the on-chip trace SRAM, and decodes the FPGA-emitted slots with the golden decoder — every clean point below round-trips bit-exact (decode=0K), confirming the HW slot stream is correct. 1-channel descriptor sweep (debug-compl monitor preset):

| records | half-beat beats | reduction                            |
|---------|-----------------|--------------------------------------|
| 4       | 12              | 0.0 % (all first-occurrence escapes) |
| 8       | 17              | 29.2 %                               |
| 16      | 23              | 52.1 %                               |
| 32      | 35              | 63.5 %                               |

The live curve tracks the offline warm-up curve (§5): reduction climbs with record count toward the full-capture 75.7 %.

**Measurement caveat (not a codec issue).** The live *multi-channel* sweep was not reproducible: 8desc\_4ch returned a CRC mismatch and 8desc\_8ch gave 405 beats one run vs 261 another for the identical config.



The cause is the known back-to-back-UART **cluster-reset fragility** (separate hb\_measure invocations wedge cluster state unless fully reset between runs), *not* the compressor — which is bit-exact in cosim and deterministic offline. Those points are excluded; the offline model (§2–5) is the authoritative source. Investigating a robust per-run cluster reset is tracked as follow-up.

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## 7. What we learned

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1. **Half-beat is worth it on real traffic:** 63.0 % → 75.7 % (+12.7 pp) on the captured stream, for two 30-bit slots per beat.
  2. **h dominates, not slot size.** At  $h = 0.944$  the half-beat ceiling is 78.6 %; we hit 75.7 % of the theoretical 83.3 %.
  3. **Three quarters of records half-pack;** 18 % need a full beat (wide payload), 6 % escape (cold templates).
  4. **Warm-up is the real-world tax.** Short streams under-report; the cost is one 3-beat escape per distinct template, amortized over the run.
  5. **The HW codec is correct.** Live FPGA slots decode bit-exact and track the model; the only live wobble is measurement-harness reset fragility, not the RTL.
- 

## Appendix: data files & reproduce

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| File                                                                                                                                                                                                                                                                                                                                                                | Contents                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| ../compression_dataset/<br>desc_axi_16desc_8ch_1MB.json                                                                                                                                                                                                                                                                                                             | real 682-record on-board capture<br>(locked dataset)           |
| plots/*.png                                                                                                                                                                                                                                                                                                                                                         | figures above<br>(host/plot_compression_report.py)             |
| plots/summary.json                                                                                                                                                                                                                                                                                                                                                  | numeric summary (records, h, tier<br>split, beats, reductions) |
| <pre>cd flows-stream-bridge/host &amp;&amp; source \$REPO_ROOT/env_python # Authoritative, offline, bit-exact (no board needed): python3 plot_compression_report.py \   --capture ../reports/compression_dataset/desc_axi_16desc_8ch_1MB.json \   --outdir ../reports/compression/plots # Live on-board confirmation (keep workloads small; trace SRAM = 2048</pre> |                                                                |

```
beats;  
# reset the cluster between runs):  
python3 hb_measure.py --port /dev/ttyUSB2 --channels 1 \  
    --config 8desc_1ch_1MB --mon-config debug-compl
```

Codec spec: bin/TBClasses/monbus/monbus\_compressor.py; RTL:  
rtl/amba/shared/monbus\_{compressor,halfbeat\_packer}.sv; cosim:  
val/amba/test\_monbus\_{compressor,halfbeat}.py.